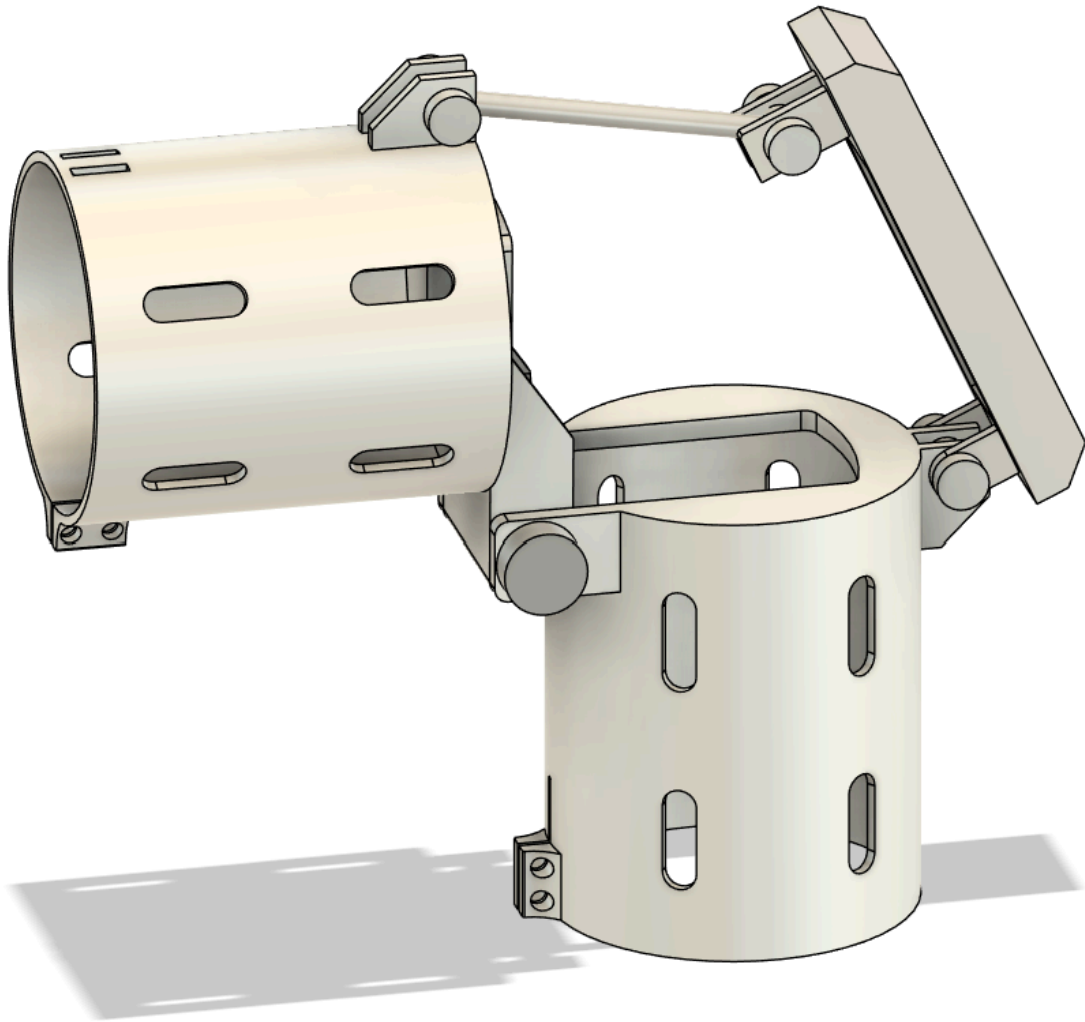


Quick-Release Folding Arm Joint for UAV Motor Arm



By Sreeram Padmesh

Executive Summary

Concept Overview:

This document will explain the whole design and analysis of the quick release folding arm joint intended for a 50mm arm tube and 60kg in-flight load. The joint is designed to allow a sideways folding action and maintaining high rigidity when locked into the extended, operational position.

Its a simple interference fit plus magnetic stop lock system, that can be pulled out and folded as require and can be kept back into operational form by unfolding and clicking it back into place.

Key Design Choices:

- Wall thickness of 1.5mm, reduces system weight while the Aluminium 7075 material ensures the tube's structural integrity
- Static Load = $60\text{kg} \times 1.98\text{m/s} = 588.6\text{N}$, with a 2g load factor, giving a design load of $2 \times 588.6 = 1177.2\text{N}$
- Safety factor of 2.5 is a sustainable value, between both safety and mass efficiency
- Must withstand $1177.2 \times 2.5 = 2943\text{N}$
- Materials used Aluminium 7075, Stainless Steel and PTFE

Weight Estimate:

Taken from Fusion360's properties tab for the full assembly after having defined all the materials for the CAD model, the estimated weight of the product is 166g

Vibration Damping Strategy:

Precision while manufacturing is one of the most important methods for vibration damping and to get such precision using manufacturing processes like CNC machining is very crucial.

Additionally adding layers of elastomer gaskets between metal on metal contact regions which bear most of the load will damp the vibration by converting the high frequency energy into heat.

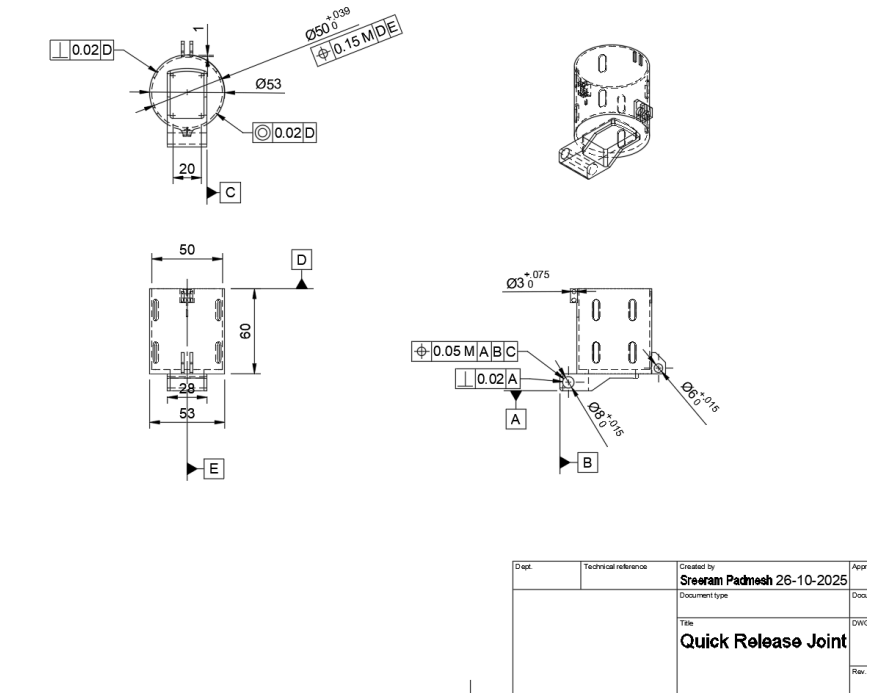
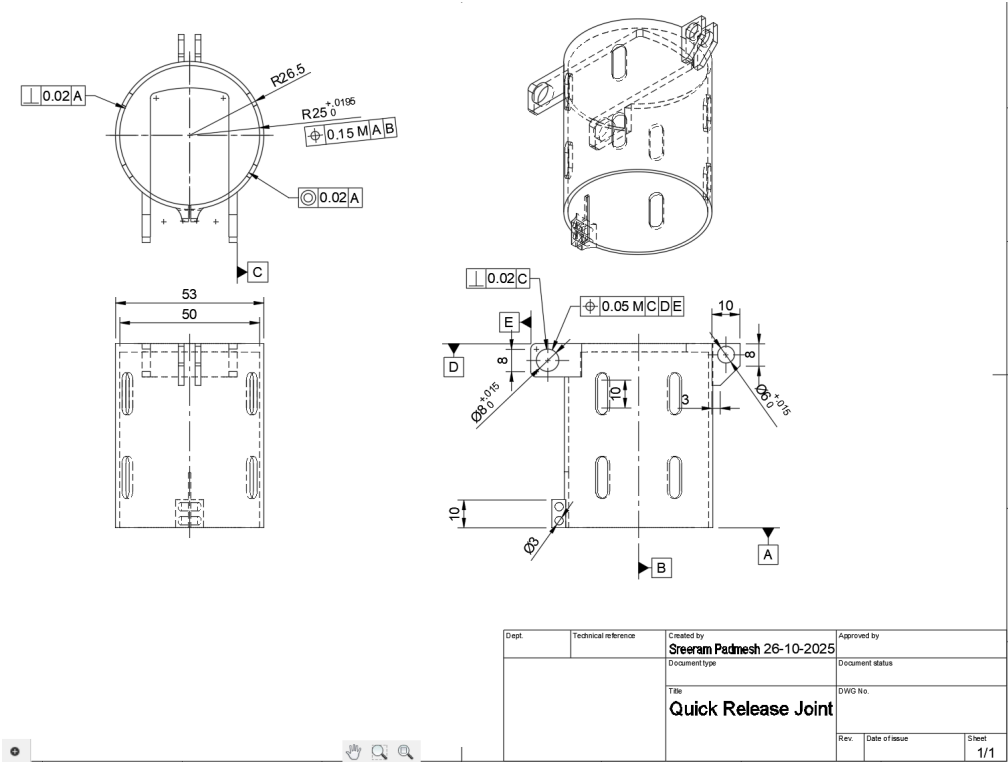
CAD Model

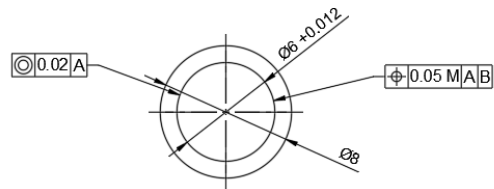
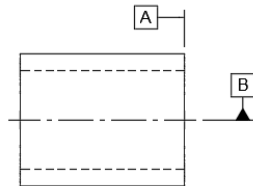
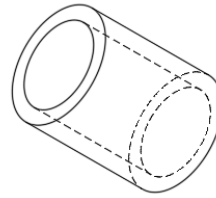
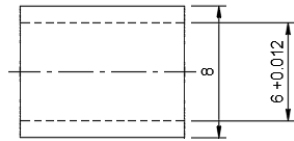
Assembly Instructions:

There are totally four main parts, the two joint blocks that fold and the two links that connect the two blocks together on the opposite side of the hinge.

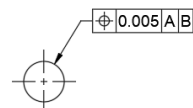
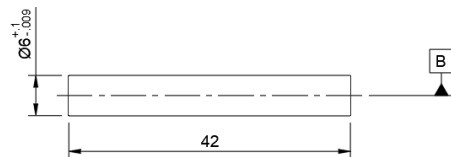
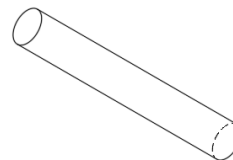
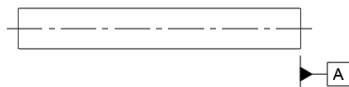
1. The Stainless Steel Hinge Pin (8mm) is inserted through the bore of the **static hinge block** passing through the internal bushings mounted in the **rotating hinge block** and exiting the far side of the two joint blocks.
2. Low friction **PTFE Bearings** are pressed into the blocks between the hinge pin and the blocks.
3. **Stainless Steel Insert Lock Nuts** are threaded onto the pin at both ends. The nuts are tightened to achieve a defined **axial preload**.
4. There is a **Stainless Steel Rod**, acts as one of the links which is slotted into two drilled pins (6mm) tightly.
5. The pins have been inserted like through at the opposite end of the hinge on three separate locations, to connect static block to **PTFE Link**, rotating block to stainless steel rod and the PTFE link to the stainless steel rod
6. **Stainless Steel Insert Lock Nuts** are threaded onto the pins at both ends same way as step 3
7. While inserting 50mm arm tube, there are 3mm holes through which you may insert a nut and bolt to tighten the grasp of the arm tubes

2D Drawings - Joint Blocks, Bearing and Hinge Pin





Dept.	Technical reference	Created by Sreeram Padmesh 28-10-2025	Approved by
		Document type	Document status
		Title Quick Release Joint	DWG No.
		Rev.	Date of issue
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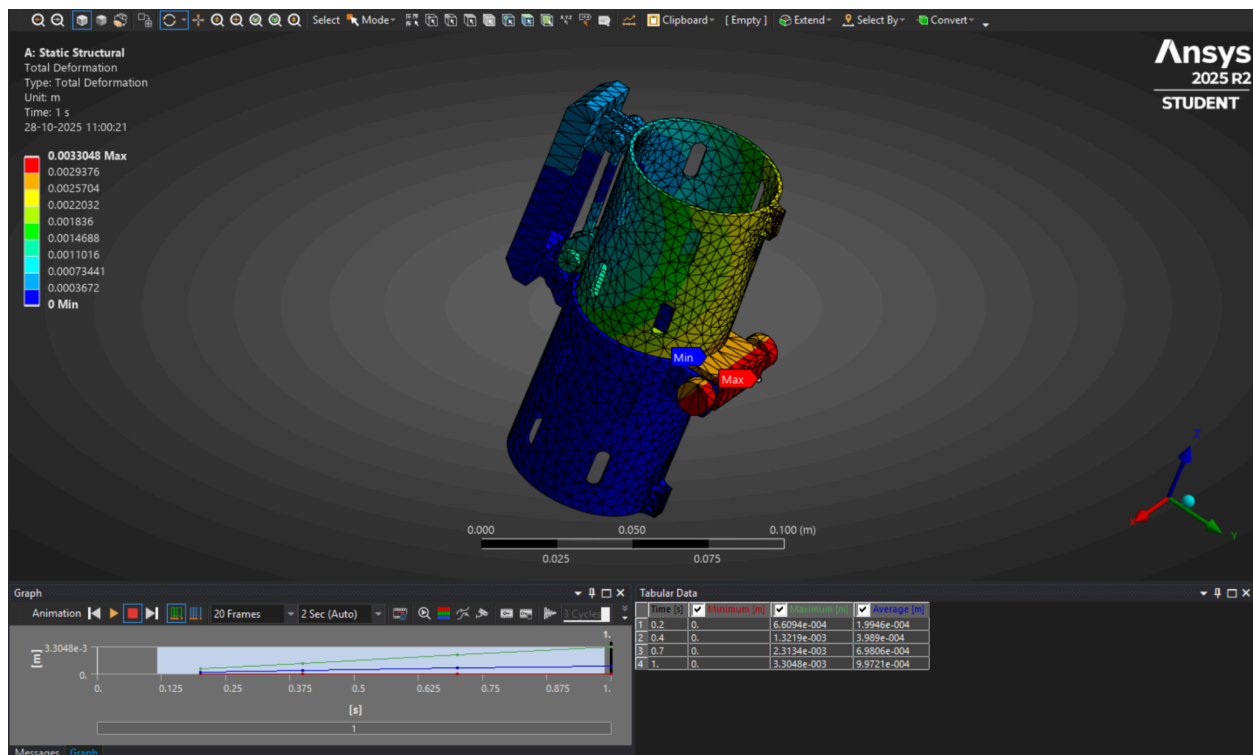
FEA Report

Static Structural:

- Bonded joints everywhere except the hinge at which there are frictionless contact regions between the pin and bearing
- Fixed support for one of the joint blocks while the other carries a quarter of the 60kg in flight load as assumed for a quadcopter
- 5mm meshing size and quadratic element order for the assembly
- 2mm meshing size for the hinge pin in contact with bearing
- 2mm meshing size for the bearing in contact with hinge pin
- 4mm meshing size at locking area to get a better result of low safety factor seen at the point

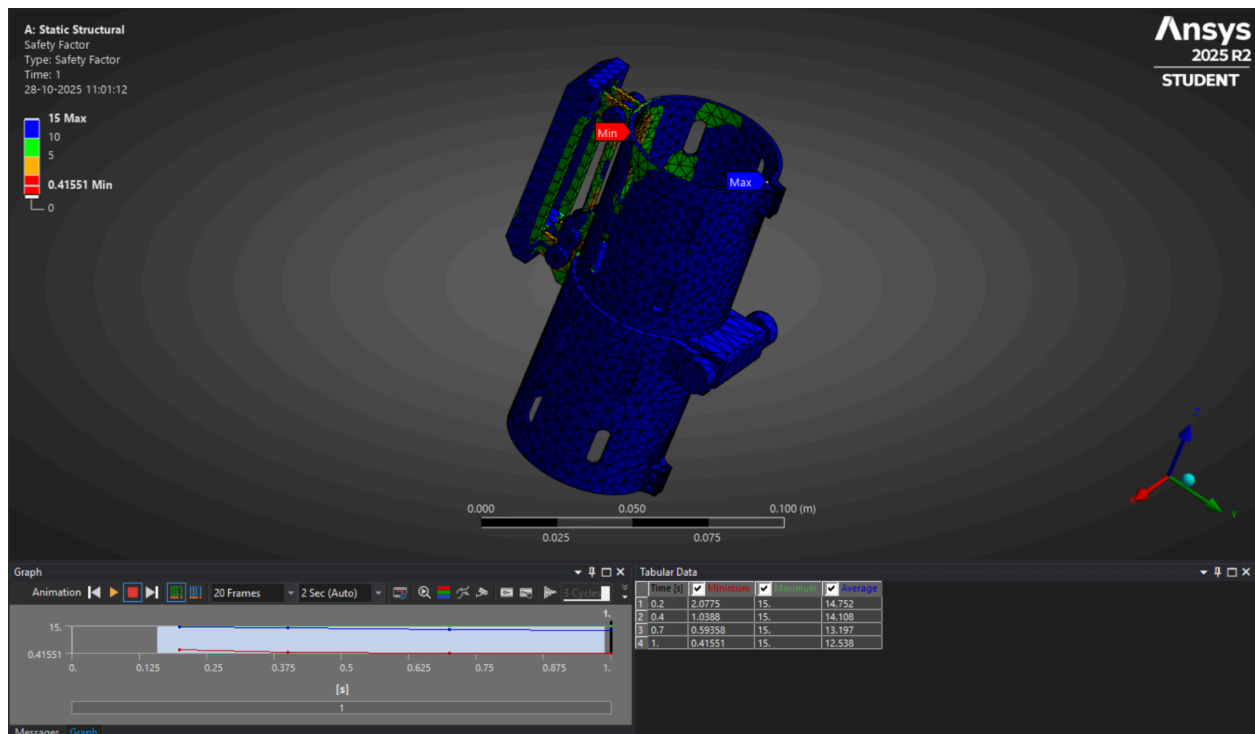
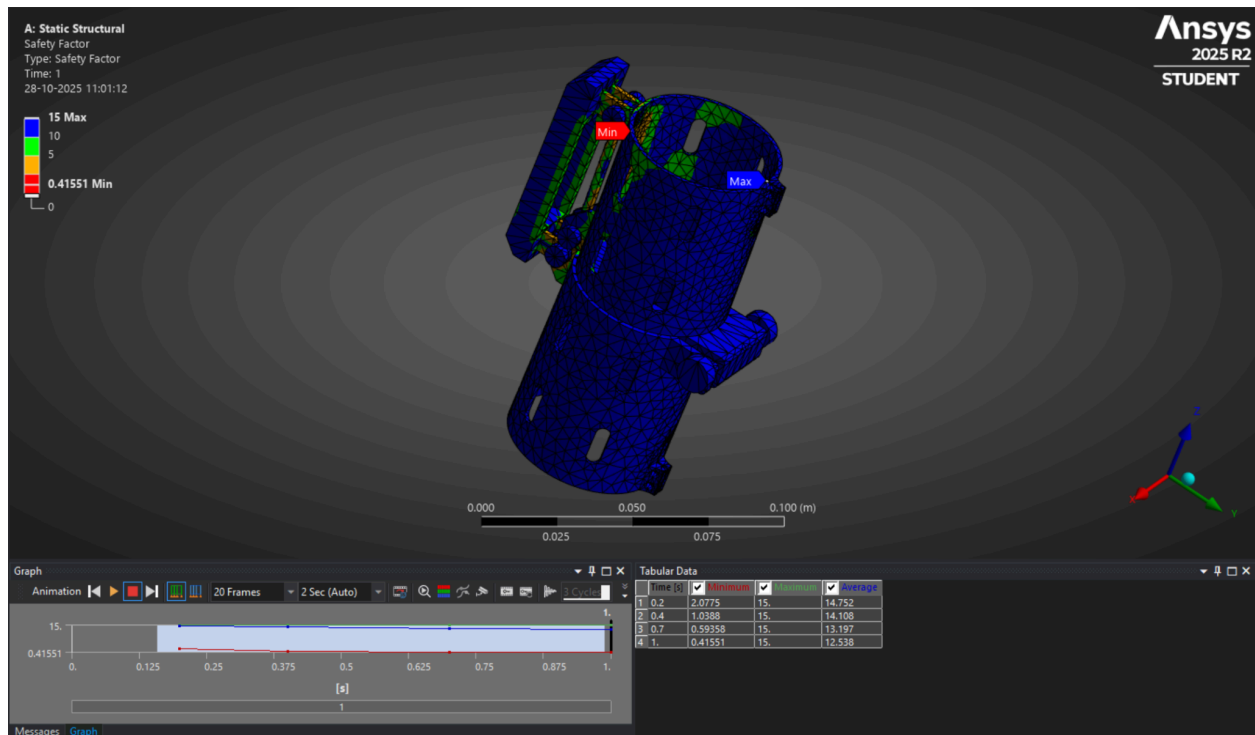
Static Structural - Total Deformation

Deformation is higher at the hinges with a value of about 3mm meaning if given a 100kg load to a drone, 981N by 4 legs for a quadcopter and a 2g dynamic factor, a 490.5N load only deforms the joint maximum by 3mm



Static Structural - Equivalent Stress and Safety Factor

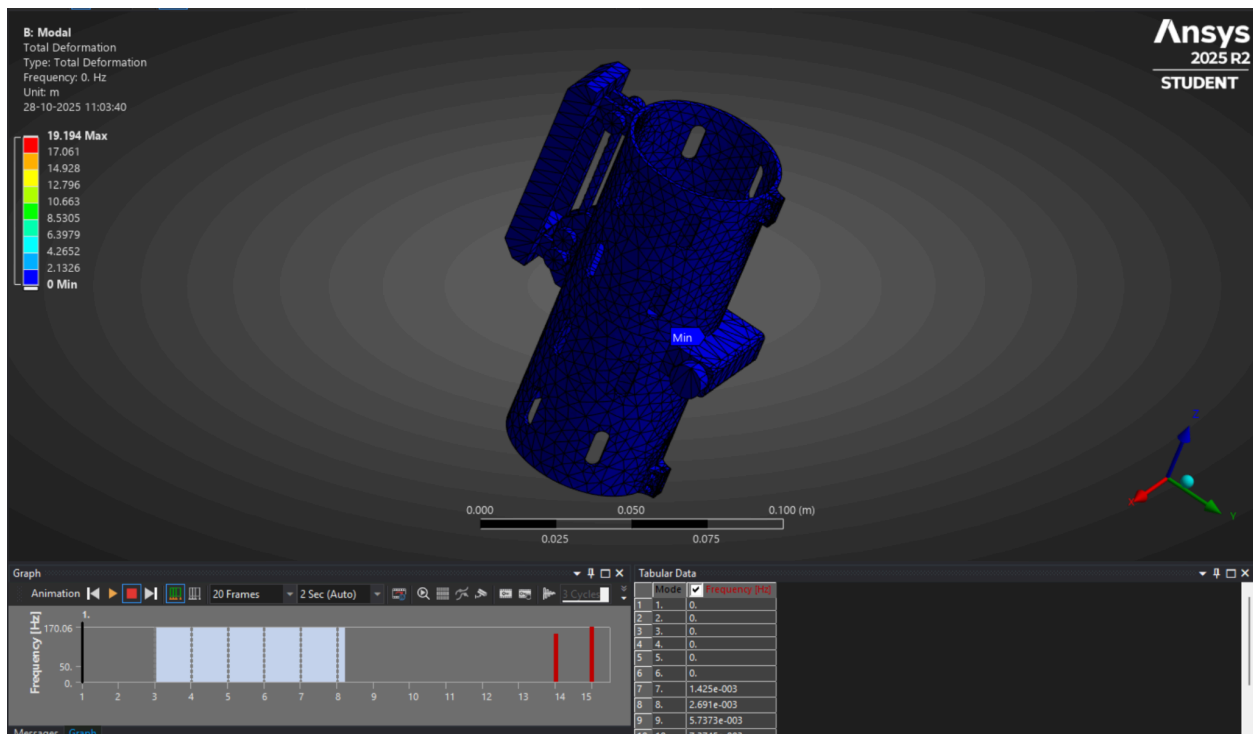
The stresses are very high at the locking mechanism which I gave to be a simple interference fit with a magnetic lock mechanism. This mechanism needs to be changed to something like a CAM clamp mechanism for better results. Everything is fine except the area of locking which needs to be addressed.



Modal Analysis

- Bonded joints everywhere except the hinge at which there are frictionless contact regions between the pin and bearing
- Fixed support for one of the joint blocks while the other carries a quarter of the 60kg in flight load as assumed for a quadcopter
- 5mm meshing size and quadratic element order for the assembly
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There are no problems as per the frequency table as it is very very low



Bill Of Materials (BOM)

Parts	Number of Parts	Materials	Mass (g)	Cost (Rs)
Joint Block	2 (Differ in geometry)	Aluminium 7075	67.5 and 52.4	$2000 \times 2 = 4000$
Pin	4 (Differ in length and diameter)	Stainless Steel	9.3, 2.9 (2), 3.5	$200 \times 4 = 800$
Nut	8 (Differ in length and diameter)	Stainless Steel	0.6 (6), 4.4 (2)	$20 \times 8 = 160$
Rod	1	Stainless Steel	3.8	200
Link	1	PTFE	10	300
Bearing	2	PTFE	0.5 (2)	$300 \times 2 = 600$

Manufacturing and Assembly Notes

Manufacturing Processes

Joint blocks will be manufactured by 5-Axis CNC Milling as it has a complex geometry, require tight tolerances. Hard Anodization (Type III) on the 7075 blocks is non-negotiable for corrosion protection and surface hardening, especially since 7075 is prone to Stress Corrosion Cracking.

Hinge Pins will be manufactured using CNC turning and grinding that ensures we get the geometry and threading required for its purpose. A medium-strength thread-locking compound (e.g., Loctite 243) must be

applied to the threads of the hinge pin before the final locknuts are installed.

Bearings will be manufactured using CNC milling which is enough to get the required shape and it retains low friction properties for pin rotation and smooth movement of the blocks

The primary fasteners are the stainless steel threaded pin and the lock nuts. These fasteners are chosen specifically because they resist loosening under high-frequency vibration and allow for precise application of axial preload.

Critical Tolerances

Tight control over tolerances is paramount for eliminating "slop" (unwanted play) and guaranteeing the structural rigidity provided by the lock-up geometry.

- **Pin/Bearing Fit:** The clearance between the stainless steel hinge pin and the bearing must be precise to allow free rotation while minimizing play. A **H7/g6 fit** is specified (typical clearance tolerance of 0.005mm to 0.020mm).
- **Mating Faces:** The aluminum blocks where the folding stop occurs, and where the nuts apply preload, must be machined flat to a tolerance of less than or equal to 0.01mm. This is essential to ensure uniform load distribution and correct axial preloading.
- **Arm Tube Fit:** The bore of the block that receives the 50mm arm tube requires an **H8 tolerance** to allow for easy assembly with an appropriate bond line.

Replacing Wear Items

The only expected wear item is the PTFE Bearing, the stainless steel pin and locknuts

1. **Disassembly:** Carefully remove the external lock nuts from the hinge pin. Just push the threaded hinge pin out of the joint blocks. Once the pin is removed, the joint blocks can be separated.
2. **Inspection and Replacement:**
PTFE Bearing: Visually inspect the PTFE component. If any

significant flattening, cracking, or loss of material is observed, replace it with a new, precision-machined unit.

Hinge Pin: Inspect the stainless steel pin for scoring or significant surface wear.

Lock Nuts: Always use new Lock Nuts if the old ones have been removed more than once, as repeated use compromises their vibration resistance.

3. **Reassembly:** Insert the new PTFE bushing into the block. Align the two joint blocks and insert the hinge pin. Thread the new lock nuts onto both ends. Tighten the nuts carefully to apply a controlled axial preload.

Test Plan

Static Pull/Ultimate Load Test:

This test verifies the safety margin built into the design by applying loads until failure

Procedure:

1. Mount the assembly onto a Universal Testing Machine (UTM)
2. Apply tensile load in the direction of sustained flight direction
3. The load will reach proof load, then ultimate load

Acceptance:

- Proof load > 1177.2N
- Ultimate load > 2943N
- Failure must occur above Ultimate Load

Fatigue Test:

This test validates the durability of the folding mechanism and the wear items over the product lifespan

Procedure:

1. Mount it on a testing platform which folds and unfolds the joint continuously
2. The locking mechanism has to be tested as well, so while unfolding, it must be locked into place each time

Acceptance:

- Cycle Life: The joint must successfully complete a minimum of 3,000 fold/unfold cycles
- The PTFE Bearing must be inspected. It can show wear but no signs of failure.

Vibration Testing

This verifies the integrity of the critical preloaded hinge pin assembly under the stress of continuous propeller vibration.

Procedure:

1. Place joint on a shaker table
2. At locked position it must face frequencies from 0 - 500Hz that mimic motor frequencies that it may feel in real time

Acceptance:

1. No cracks, yielding, or separation of the bonded arm tube from the 7075 block
2. Lock nuts should be checked and inspected if it can still be used

Exploded View

